

Department of Computer Science and Engineering

Individual Contribution Report — Team Assignment

To be completed by EACH student individually and submitted along with the team's assignment. This report explains what YOU specifically did, so it must be written in your own words and reflect only your own work.

A. Assignment & Team Information

Course Code & Name	CSA0614 – DESIGN AND ANALYSIS OF ALGORITHM	
Assignment Title	Conflict-Free Exam Timetable Scheduling Using Graph Colouring and Backtracking	
Team / Group No.	TEAM-09	
Team Members (all names)	192472016	KARANAM SAI DATTA
	192425306	T. Chandra Mouleswarao
	192425294	S. Mahammed Nayeem
Date of Submission	02-09-2026	

B. Student's Own Details

Student Name	S. Mahammed Nayeem
Register / Roll No.	192425294
Role in the Team	Algorithm Implementation and Graph Generation

C. Task-wise Individual Contribution

List every task/module you personally worked on. Add rows if needed. Do not describe work done by teammates.

S.No	Task / Module Handled	Description of Work Done (in your own words)	Tools / Techniques Used	Approx. Hours Spent
1.	Test Case Preparation	Prepared different test cases for evaluating the Graph Colouring algorithms with varying numbers of courses and time slots.	Python, Graph Theory, Adjacency Matrix	4 Hours
2.	Sparse Graph Testing	Conducted experiments on sparse conflict graphs containing fewer conflicts between courses to evaluate the performance of both algorithms.	Python, Recursion, Backtracking	3 Hours
3.	Dense Graph Testing	Conducted experiments on dense conflict graphs with highly overlapping course selections	Graph Colouring, Pruning, Heuristic Search	2 Hours
4.	Algorithm Validation	Measured execution time, backtracking steps, conflicts explored and memory utilisation for both the plain and pruning-enhanced Backtracking algorithms.	Testing, Constraint Validation	3 Hours

D. Team Contribution Percentage Summary

To be filled identically by every team member so contributions are consistent and sum to 100%. Disagreements should be discussed and resolved before submission.

Team Member Name	Register No.	% Contribution	Signature
T. Chandra Mouleswarao	192425306	33.33%	
S. Mahammed Nayeem	192425294	33.33%	S.Mahammed Nayeem
K. SAI DATTA	192472016	33.33%	
Total		100%	

E. Course Outcomes / Skills Applied

Course Outcome(s) Addressed by MY Work	How I Applied It
CO1 – Analyse the efficiency of recursive and non-recursive algorithms using asymptotic notations and complexity-analysis methods.	I applied recursive algorithm concepts while implementing the Backtracking approaches. I understood how the recursive search process explores different colour assignments and how the computational effort increases as the number of courses increases.
CO2 – Analyse the growth of solution-space search as problem size increases using asymptotic reasoning.	Through the implementation of the Graph Colouring algorithms, I observed how the number of possible colour assignments grows as the number of courses and available time slots increases. My implementation supported testing different graph sizes for scalability analysis.
CO3 – Apply the Backtracking technique, including pruning strategies, to solve constraint-satisfaction problems such as Graph Colouring.	I implemented both the plain Backtracking algorithm and the pruning-enhanced Backtracking algorithm for the examination scheduling problem. I applied conflict checking and heuristic-based vertex selection to reduce unnecessary exploration during the search process.

F. Challenges Faced and How I Resolved Them

Describe any technical or teamwork challenges you personally faced, and the steps you took to resolve them.

One of the major challenges I faced was ensuring a fair comparison between the plain Backtracking algorithm and the pruning-enhanced algorithm. I resolved this by testing both algorithms in the same computational environment using identical conflict graphs and the same fixed number of colours.

Another challenge was selecting appropriate graph sizes and edge densities for experimentation. I addressed this by conducting tests using both sparse and dense conflict graphs with increasing numbers of courses.

Measuring performance accurately was also challenging because execution time alone did not provide a complete comparison. Therefore, I included additional metrics such as the number of backtracking steps, conflicts explored and memory utilisation to obtain a more detailed performance evaluation.

G. Individual Learning Outcome / Reflection

What did YOU personally learn from completing your part of this assignment? How did it build on classroom concepts?

Through my contribution to this assignment, I gained practical knowledge of algorithm testing and experimental performance analysis. I learned how the characteristics of input data, particularly graph size and edge density, can significantly affect the performance of Backtracking algorithms.

I also understood the importance of using multiple performance metrics instead of relying only on execution time. Measuring backtracking steps, conflicts and memory usage provided a clearer understanding of the algorithms' scalability.

This assignment helped me connect theoretical algorithm concepts with experimental evaluation. I learned how to design fair experiments, analyse algorithm performance and evaluate the effectiveness of pruning techniques in solving constraint-satisfaction problems.

H. Declaration

I declare that the contribution described above accurately represents the work I personally performed for this team assignment, and that I have not misrepresented the work of any other team member as my own.

Student Signature	Date
S.Mahammed Nayeem	02-09-2026